

# Single-Bend Analytic Placement Tests

OPALX element placing sandbox

May 29, 2026

## 1 Purpose

The sandbox inputs

|                         |                                      |
|-------------------------|--------------------------------------|
| SBEND placement         | sandbox/test-sbend-1/test-sbend-1.in |
| SBEND vertical focusing | sandbox/test-sbend-2/test-sbend-2.in |
| RBEND placement         | sandbox/test-rbend-1/test-rbend-1.in |
| RBEND vertical focusing | sandbox/test-rbend-2/test-rbend-2.in |

exercise the OPALX-native analytic **SBEND** and **RBEND** without field maps, using the native analytic bend model and explicit poses. The first test of each bend type is a placement consistency test. The second test adds a vertical offset and checks the pole-face/fringe-field focusing against old OPAL 2022.1.

Both tests use explicit element posing:

$$(X, Y, Z, \Theta, \Phi, \Psi),$$

where  $(X, Y, Z)$  is the placed body-frame origin and  $\Theta$  is the body rotation about the global  $Y$ -axis in the  $X$ - $Z$  drawing plane. The figures use this global  $X$ - $Z$  projection; for these horizontal-bend tests the body pose has  $Y = 0$ , normal to the drawing plane.

## 2 Common analytic SBEND geometry

For both tests

$$L_{\text{chord}} = 1.0 \text{ m}, \quad \theta = 45^\circ = \frac{\pi}{4}, \quad R = \frac{L_{\text{chord}}}{2 \sin(\theta/2)} = 1.3065629649 \text{ m}.$$

For parser-created **SBEND**, OPALX follows old OPAL and treats  $L$  as the chord. The placed sector arc length and curvature are therefore

$$L_{\text{ref}} = R\theta = 1.0261721530 \text{ m}, \quad h = \frac{\theta}{L_{\text{ref}}} = \frac{1}{R} = 0.7653668647 \text{ m}^{-1}.$$

OPALX's **PlanarArcGeometry** places the body-frame origin at the arc midpoint, so the body coordinate  $s$  runs from  $-L_{\text{ref}}/2$  to  $+L_{\text{ref}}/2$ . The local sector orbit is the standard sector-bend reference-orbit convention used in first-order beam dynamics [1, 4]:

$$\mathbf{r}(s) = \begin{pmatrix} \frac{\cos(hs) - 1}{h} \\ 0 \\ \frac{\sin(hs)}{h} \end{pmatrix}, \quad \mathbf{t}(s) = \begin{pmatrix} -\sin(hs) \\ 0 \\ \cos(hs) \end{pmatrix}.$$

For  $L_{\text{chord}} = 1$  m and  $\theta = 45^\circ$ ,

$$\mathbf{r}_{\text{entry}} = (-0.0994561837, 0, -0.5000000000) \text{ m}, \quad \mathbf{r}_{\text{exit}} = (-0.0994561837, 0, 0.5000000000) \text{ m}.$$

The longitudinal field support follows OPAL's default `FM1DProfile1` Enge profile. With no explicit GAP, the inputs use  $g = 2 \text{ HGAP} = 0.04$  m, so the default Enge half width is  $a = 5g = 0.2$  m:

$$\ell_i = \frac{a}{|\cos E_i|}, \quad L_{\text{eff}} = \int_{-\ell_1}^{L_{\text{ref}} + \ell_2} F_{\text{Enge}}(s) ds.$$

The vertical pole-face correction used for the thin-edge estimate is

$$\psi(E) = h_{\text{eff}} \text{HGAP} F_{\text{int}} \frac{1 + \sin^2 E}{\cos E}, \quad k_y(E) = -h_{\text{eff}} \tan(E - \psi(E)).$$

The thin-entry-edge vertical map used to design the second test is

$$y_{\text{exit}} = (1 + L_{\text{ref}} k_y(E_1)) y_{\text{entry}} + L_{\text{ref}} y'_{\text{entry}}.$$

### 3 test-sbend-1: placement consistency test

This test checks the first explicitly posed analytic `SBEND` pole-face case: a  $45^\circ$  entrance face, zero exit face, a 0.3 m upstream drift, and the initial particle exactly at  $(0, 0, 0)$ . Because the particle is on the reference point, the entrance pole face should not deflect it away from the design exit. The exit monitor shown in Fig. 1 is placed at the exported `B1` exit edge. The input deck derives the body and monitor poses from the same sector geometry used by old OPAL. With

$$\mathbf{r}_0 = (X_0, 0, Z_0), \quad X_0 = 0, \quad Z_0 = d, \quad d = 0.3 \text{ m},$$

the formulas used in `test-sbend-1.in` are

$$\begin{aligned} R &= \frac{L_{\text{chord}}}{2 \sin(\theta/2)}, \\ \mathbf{r}_{\text{body}} &= \mathbf{r}_0 + \begin{pmatrix} R(\cos(\theta/2) - 1) \\ 0 \\ R \sin(\theta/2) \end{pmatrix}, \quad \Theta_{\text{body}} = -\frac{\theta}{2}, \\ \mathbf{r}_{\text{exit}} &= \mathbf{r}_0 + \begin{pmatrix} R(\cos \theta - 1) \\ 0 \\ R \sin \theta \end{pmatrix}, \quad \Theta_{\text{exit}} = -\theta, \\ \mathbf{t}_{\text{exit}} &= \begin{pmatrix} -\sin \theta \\ 0 \\ \cos \theta \end{pmatrix}. \end{aligned}$$

For the present  $45^\circ$  case this gives

$$\begin{aligned} \mathbf{r}_{\text{entry}} &= (0, 0, 0.3000000000) \text{ m}, \\ \mathbf{r}_{\text{body}} &= (-0.0994561837, 0, 0.8000000000) \text{ m}, \\ \mathbf{r}_{\text{exit}} &= (-0.3826834324, 0, 1.2238795325) \text{ m}, \\ \Theta_{\text{body}} &= -22.5^\circ, \\ \Theta_{\text{exit}} &= -45^\circ. \end{aligned}$$

| quantity          | value                                                              |
|-------------------|--------------------------------------------------------------------|
| ANGLE             | $45^\circ$                                                         |
| E1, E2            | $45^\circ, 0^\circ$                                                |
| HGAP, FINT        | 0.02 m, 0.5                                                        |
| $\ell_1, \ell_2$  | 0.2828427125, 0.2000000000 m                                       |
| $L_{\text{ref}}$  | 1.0261721530 m                                                     |
| $L_{\text{eff}}$  | 1.0261728620 m                                                     |
| $h_{\text{eff}}$  | $0.7653663359 \text{ m}^{-1}$                                      |
| SBEND body pose   | $X = -0.0994561837, Y = 0, Z = 0.8000000000, \Theta = -22.5^\circ$ |
| exit monitor pose | $X = -0.3826834324, Z = 1.2238795325, \Theta = -45^\circ$          |

The practical placement criterion is based on the monitor local coordinates:

$$r_\perp = \sqrt{\bar{x}^2 + \bar{s}^2} < 5 \times 10^{-5} \text{ m.}$$

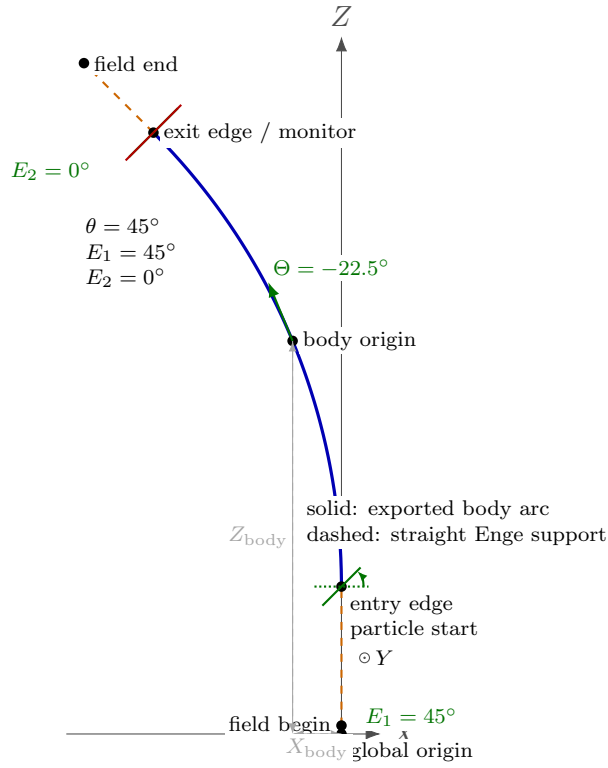


Figure 1: `test-sbend-1`: placed body arc, field support, and temporal monitor plane. The grey dimensions mark the OPALX body pose in the global  $X$ - $Z$  projection; the circled-dot axis marks global  $+Y$  normal to the page. The entrance pole face is  $45^\circ$ , but the on-axis start particle remains on the design exit.

## 4 `test-sbend-1`: OPAL 2022.1 comparison

The old-OPAL comparison uses `test-sbend-1.opal.in` with the same SBEND length, angle, pole-face angles, gap, and fringe integral. The comparison script runs OPALX and OPAL 2022.1 in separate work

directories and compares only the `DesignPath.dat` and `.stat` diagnostics; HDF5 and load-balance files are ignored.

In the 30 cm drift comparison used for these figures, the reference-particle  $B_y$  peak agrees exactly at the sampled precision. The `.stat` reference-field support starts at the same longitudinal coordinate and ends within  $3 \times 10^{-4}$  m. The remaining path differences are small compared with the bend scale: the maximum design-path differences are  $4.4 \times 10^{-4}$  m in  $x$  and  $2.1 \times 10^{-4}$  m in relative  $z$ .

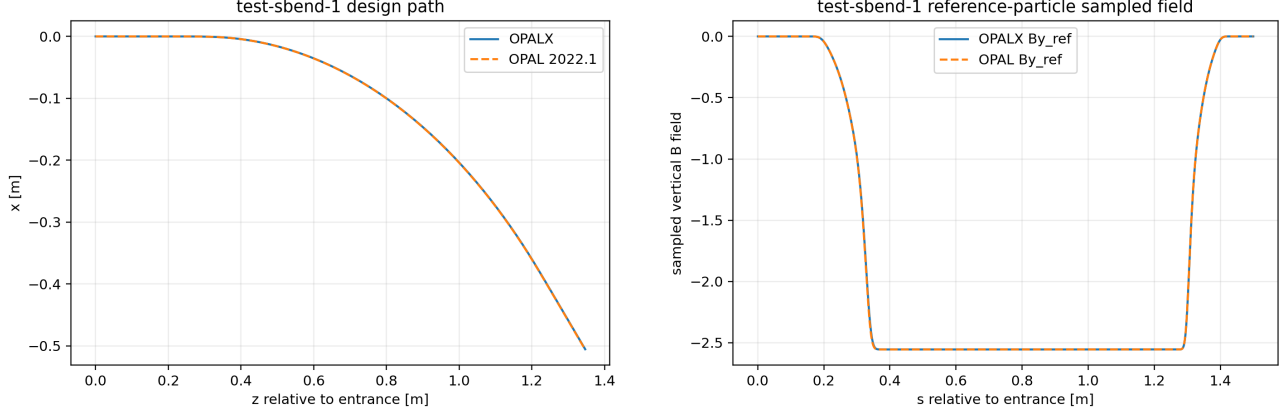


Figure 2: **test-sbend-1** OPALX versus OPAL 2022.1: design-path  $X$ - $Z$  orbit (left) and `.stat` reference  $B_y$  (right).

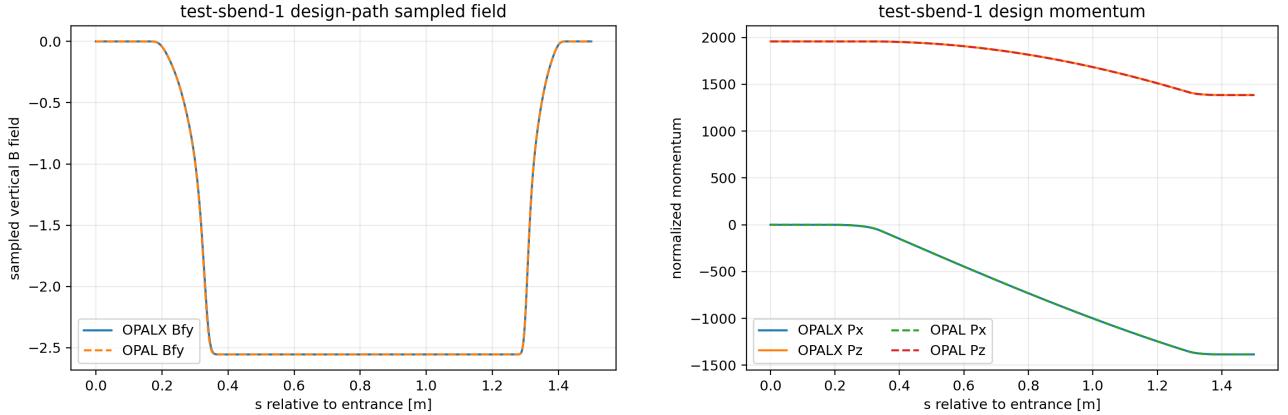


Figure 3: **test-sbend-1** design-path diagnostic comparison. Left: sampled  $B_y$  from `DesignPath.dat`. Right: design-path momentum components.

## 5 test-sbend-2: OPALX/OPAL vertical edge-focusing comparison

This test keeps the same chord-to-arc SBEND placement formulas as **test-sbend-1**, but launches a particle with

$$y_0 = 1.0 \text{ mm}, \quad y'_0 = 0.$$

The OPALX input is `sandbox/test-sbend-2/test-sbend-2.in`; the old OPAL reference input is `sandbox/test-sbend-2/test-sbend-2.opal.in`. OPALX uses explicit element placement, while old OPAL uses `ELEMEDGE`. The old OPAL distribution is replicated to  $10^4$  identical macro-particles so that OPAL 2022.1 writes the standard `.stat` diagnostics; OPALX uses a single particle and no self field.

| quantity                             | value          |
|--------------------------------------|----------------|
| upstream drift                       | 0.4 m          |
| ANGLE                                | 45°            |
| E1, E2                               | −49.99°, 0°    |
| HGAP, FINT                           | 0.02 m, 0.5    |
| OPAL full GAP                        | 0.04 m         |
| Enge half width $a = 5$ GAP          | 0.2 m          |
| $\ell_{\text{entry}} = a/ \cos E_1 $ | 0.3110798769 m |
| $\ell_{\text{exit}} = a/\cos E_2$    | 0.2 m          |
| upstream Enge support start          | 0.0889199346 m |
| nominal entrance edge                | 0.4 m          |
| nominal exit edge                    | 1.4261721530 m |

The comparison run uses

$$s_{\text{rel}}^{\text{OPAL}} = s^{\text{OPAL}} - 1 \text{ m}, \quad z_{\text{rel}}^{\text{OPAL}} = z^{\text{OPAL}} - 1 \text{ m}.$$

The apparent bend start near 0.2 m is not a different OPALX element placement. It is the visible part of the upstream Enge fringe. The full OPAL-style support begins at  $0.4 - \ell_{\text{entry}} = 0.0889199346$  m, where the field is still numerically tiny, and the nominal bend entrance remains at 0.4 m in both OPALX and old OPAL. With the comparison script's  $10^{-6}$  peak-field support threshold, the design-path  $B_y$  fringe starts at 0.2401337 m in OPALX and 0.2399838 m in old OPAL.

The reference geometry and field profile agree at the same level as the placement test:  $B_y$  peak differences are zero in the sampled output, the design-path  $B_y$  integral differs by  $1.12 \times 10^{-3}$ , and the `.stat` reference-particle field support ends within  $6.0 \times 10^{-4}$  m. The OPALX vertical edge force is now weighted with the same Enge derivative as the fringe field, so the equivalent kick is applied at the same longitudinal centroid as old OPAL's map. At the final common sample OPALX has  $\bar{y} = 2.168092 \times 10^{-3}$  m, while old OPAL has  $\bar{y} = 2.161588 \times 10^{-3}$  m. The remaining final difference is  $6.50 \times 10^{-6}$  m, so this case is now a useful vertical-focusing regression diagnostic.

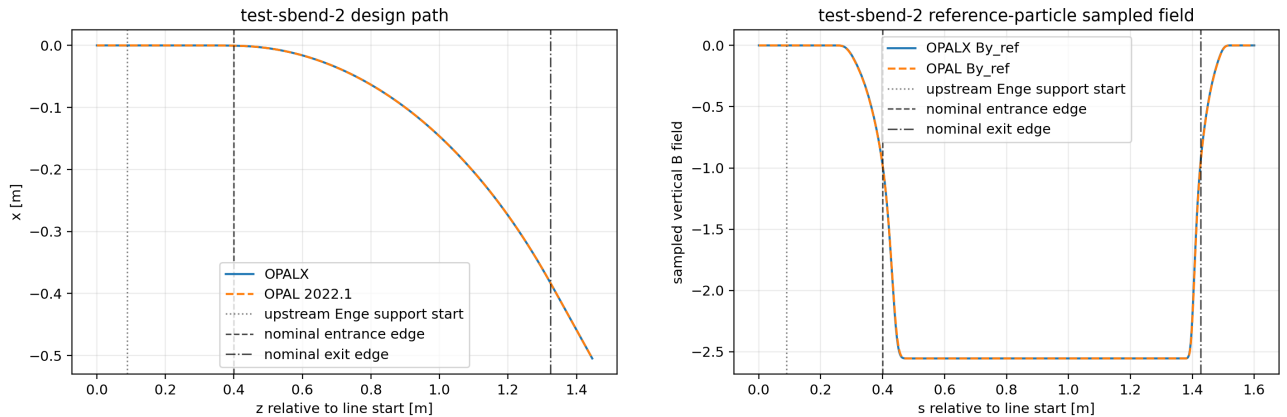


Figure 4: `test-sbend-2` OPALX versus OPAL 2022.1: design-path X-Z orbit (left) and `.stat` reference  $B_y$  (right). The dotted vertical marker is the upstream Enge support start; dashed and dash-dotted markers are the nominal entrance and exit edges.

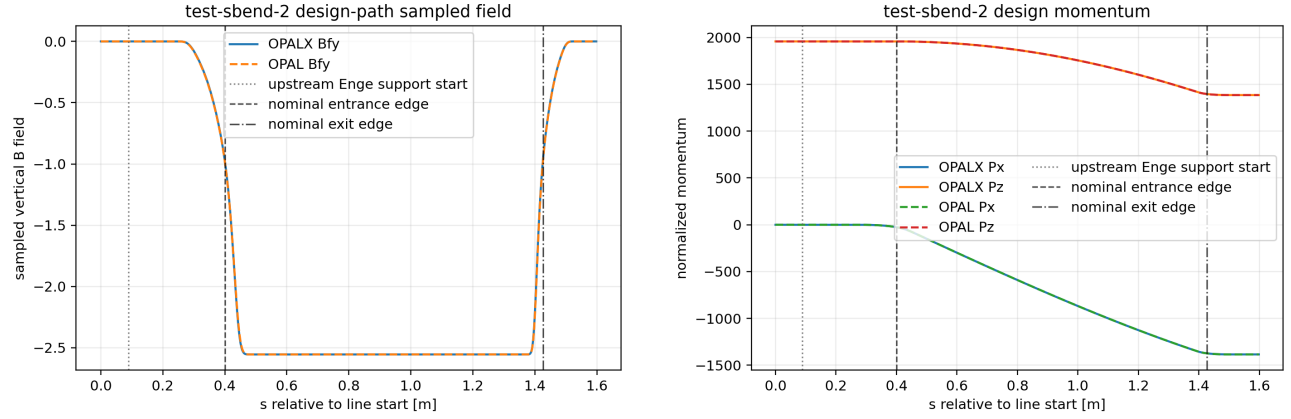


Figure 5: **test-sbend-2** design-path diagnostic comparison. Left: sampled  $B_y$  from `DesignPath.dat`. Right: design-path momentum components. The same geometry markers show that the visible fringe precedes the nominal 0.4 m entrance edge in both codes.

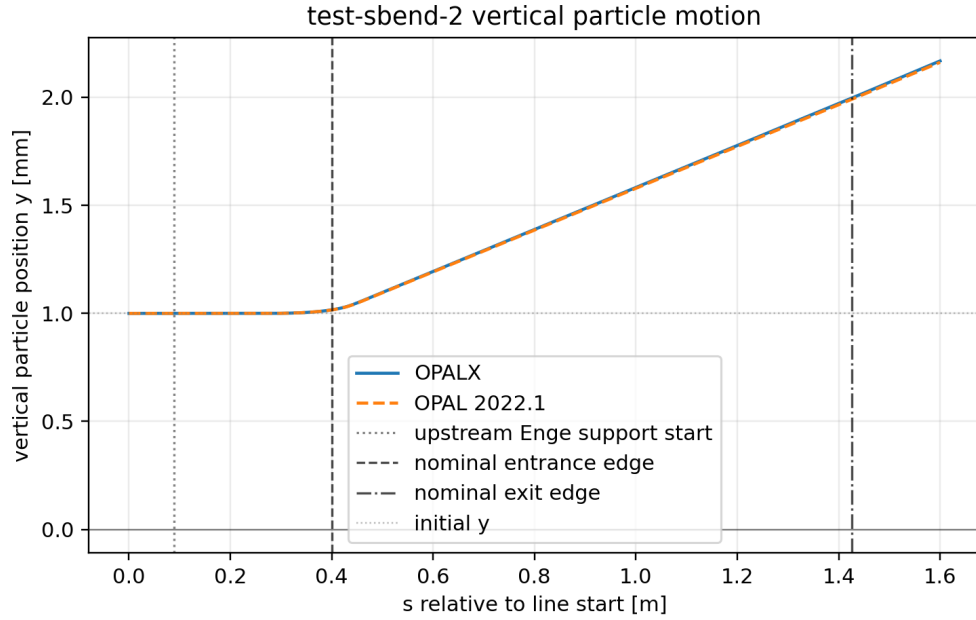


Figure 6: **test-sbend-2** vertical particle motion from the `.stat  $\bar{y}$`  column. Both distributions are monoenergetic replicated particles, so  $\bar{y}$  is the tracked particle's vertical coordinate.

## 6 Common analytic RBEND geometry

The **RBEND** tests use the same bend angle and field model as the **SBEND** tests, but the geometry is rectangular. The input length is the straight rectangular body length,

$$L_{\text{rect}} = 1.0 \text{ m}, \quad \theta = 45^\circ, \quad E_1 = 0, \quad E_2 = \theta.$$

The reference trajectory is still compared with a circular arc. For a rectangular bend the arc radius and arc length are derived from the straight body length,

$$R = \frac{L_{\text{rect}}}{2 \sin(\theta/2)}, \quad L_{\text{ref}} = R\theta = L_{\text{rect}} \frac{\theta/2}{\sin(\theta/2)} = 1.0261721530 \text{ m}.$$

The OPALX body pose is the rectangular body frame. With the requested entrance edge at

$$\mathbf{r}_0 = (X_0, 0, Z_0), \quad X_0 = 0, \quad Z_0 = d,$$

the body frame is rotated by

$$\Theta_{\text{body}} = -\frac{\theta}{2}.$$

The local coordinate of the entrance edge relative to the rectangular body origin is

$$\mathbf{q}_{\text{entry}} = \begin{pmatrix} -\frac{L_{\text{rect}}}{2} \tan(\theta/4) \\ 0 \\ \frac{L_{\text{rect}}}{2} \end{pmatrix}.$$

Using the same  $X$ - $Z$  rotation convention as the OPALX input files,

$$\mathcal{R}_Y(\Theta) \begin{pmatrix} x \\ 0 \\ z \end{pmatrix} = \begin{pmatrix} \cos \Theta x + \sin \Theta z \\ 0 \\ -\sin \Theta x + \cos \Theta z \end{pmatrix},$$

the explicit body pose and exit monitor are

$$\begin{aligned} \mathbf{r}_{\text{body}} &= \mathbf{r}_0 - \mathcal{R}_Y(\Theta_{\text{body}}) \mathbf{q}_{\text{entry}}, & \Theta_{\text{body}} &= -\frac{\theta}{2}, \\ \mathbf{r}_{\text{exit}} &= \mathbf{r}_0 + \begin{pmatrix} -L_{\text{rect}} \tan(\theta/2) \\ 0 \\ L_{\text{rect}} \end{pmatrix}, & \Theta_{\text{exit}} &= -\theta. \end{aligned}$$

For the present  $45^\circ$  geometry,

$$\mathcal{R}_Y(-22.5^\circ) \mathbf{q}_{\text{entry}} = (-0.2832272487, 0, 0.4238795325) \text{ m},$$

so the body pose is shifted upstream of the entrance edge even though the nominal entrance edge itself remains at  $z = d$ .

## 7 test-rbend-1: placement consistency test

This test is the RBEND analogue of **test-sbend-1**: a single on-axis particle, a 0.3 m upstream drift, and an explicit RBEND body pose derived from the formulas above. The input decks are

$$\text{sandbox/test-rbend-1/test-rbend-1.in}, \quad \text{sandbox/test-rbend-1/test-rbend-1_opal.in}.$$

The exported OPALX element positions confirm the requested rectangular geometry:

| quantity      | OPALX export                                                      |
|---------------|-------------------------------------------------------------------|
| entrance edge | $X = 5.6 \times 10^{-17}$ , $Z = 0.3000000000$ m                  |
| body pose     | $X = 0.2832272487$ , $Z = -0.1238795325$ , $\Theta = -22.5^\circ$ |
| exit edge     | $X = -0.4142135624$ , $Z = 1.3000000000$ m                        |
| field support | $Z = 0.1$ to $1.609014865$ m                                      |

The placement export is therefore consistent with the analytic rectangular survey. The physics comparison is not yet at the SBEND level. In the `DesignPath.dat` comparison the largest orbit differences are  $6.09 \times 10^{-2}$  m in  $X$  and  $4.97 \times 10^{-2}$  m in relative  $Z$ . The sampled  $B_y$  peak is 4.749971 in OPALX and 2.359859 in OPAL, and the visible field support starts at  $s_{\text{rel}} = 0$  in OPALX but near 0.201 m in OPAL. This points to the RBEND field/local-coordinate evaluation, not the exported element placement, as the remaining issue.

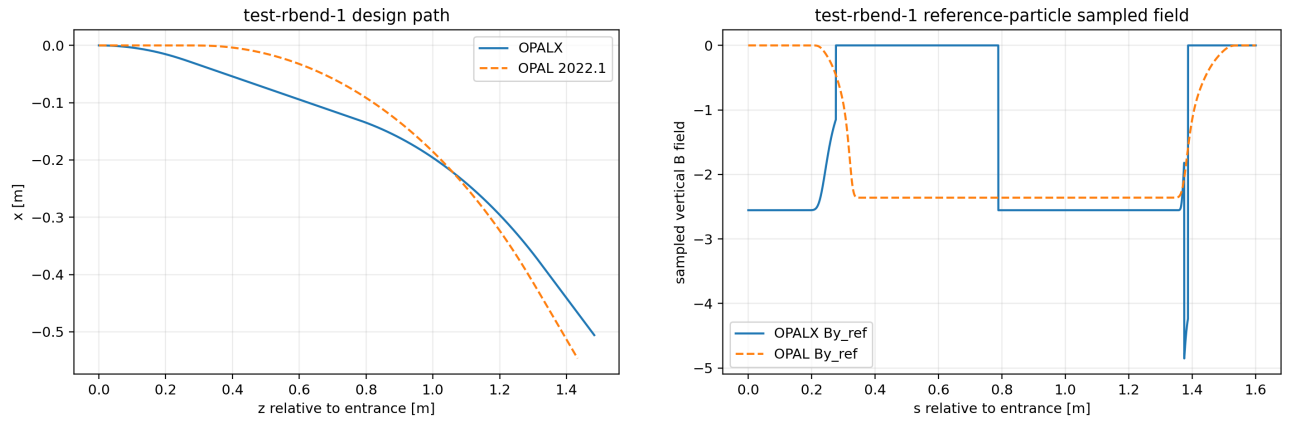


Figure 7: `test-rbend-1` OPALX versus OPAL 2022.1: design-path  $X$ - $Z$  orbit (left) and `.stat` reference  $B_y$  (right).

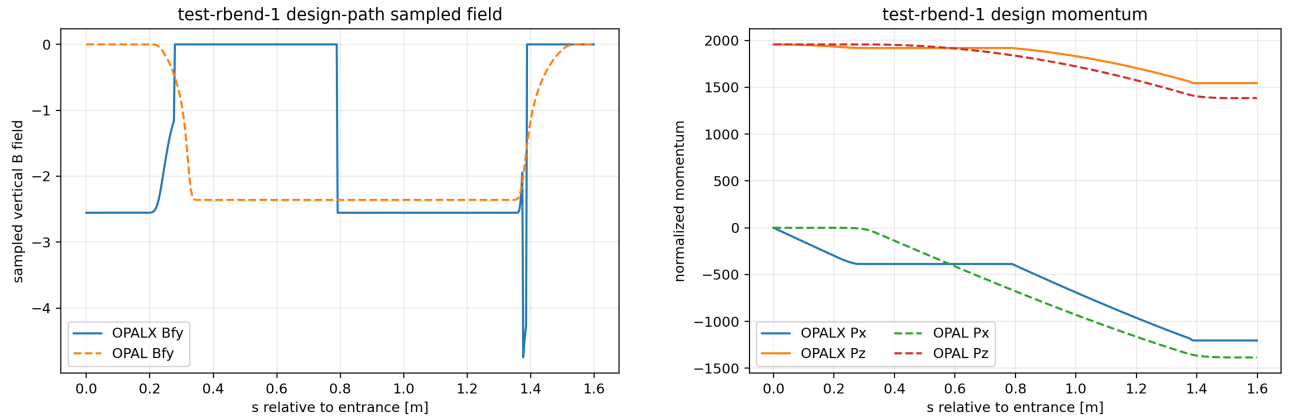


Figure 8: `test-rbend-1` design-path diagnostic comparison. The RBEND  $B_y$  peak is too large in OPALX and the momentum components diverge from old OPAL across the bend.

## 8 test-rbend-2: OPALX/OPAL vertical comparison

This test is the RBEND counterpart of `test-sbend-2`. It uses a 0.4 m upstream drift and launches the particle with

$$y_0 = 1.0 \text{ mm}, \quad y'_0 = 0.$$

The OPALX input is `sandbox/test-rbend-2/test-rbend-2.in`; the old OPAL reference input is `sandbox/test-rbend-2/test-rbend-2_opal.in`. The same rectangular formulas give

$$\begin{aligned} \mathbf{r}_{\text{entry}} &= (0, 0, 0.4000000000) \text{ m}, \\ \mathbf{r}_{\text{body}} &= (0.2832272487, 0, -0.0238795325) \text{ m}, \\ \mathbf{r}_{\text{exit}} &= (-0.4142135624, 0, 1.4000000000) \text{ m}. \end{aligned}$$

The comparison markers are

|                             |                              |                                |
|-----------------------------|------------------------------|--------------------------------|
| upstream Enge support start | $s_{\text{rel}} = 0.2000000$ | $z_{\text{rel}} = 0.2000000$ , |
| nominal entrance edge       | $s_{\text{rel}} = 0.4000000$ | $z_{\text{rel}} = 0.4000000$ , |
| nominal exit edge           | $s_{\text{rel}} = 1.4261722$ | $z_{\text{rel}} = 1.4000000$ , |
| downstream support end      | $s_{\text{rel}} = 1.7090149$ | $z_{\text{rel}} = 1.6000000$ . |

The current RBEND vertical test does not yet provide a clean edge-focusing regression, because the field discrepancy dominates the vertical dynamics. The `DesignPath.dat`  $B_y$  peak is 4.216923 in OPALX and 2.359859 in OPAL; the `.stat` reference-particle peak is 4.402652 in OPALX and 2.359859 in OPAL. At the final common `.stat` sample the mean vertical positions are

$$\bar{y}_{\text{OPALX}} = 8.094384 \times 10^{-4} \text{ m}, \quad \bar{y}_{\text{OPAL}} = 8.804121 \times 10^{-4} \text{ m},$$

for a final difference of  $-7.10 \times 10^{-5}$  m. This case is therefore useful as a diagnostic for the remaining RBEND field-coordinate problem, but it should not be promoted to a strict regression until the reference field agrees with old OPAL.

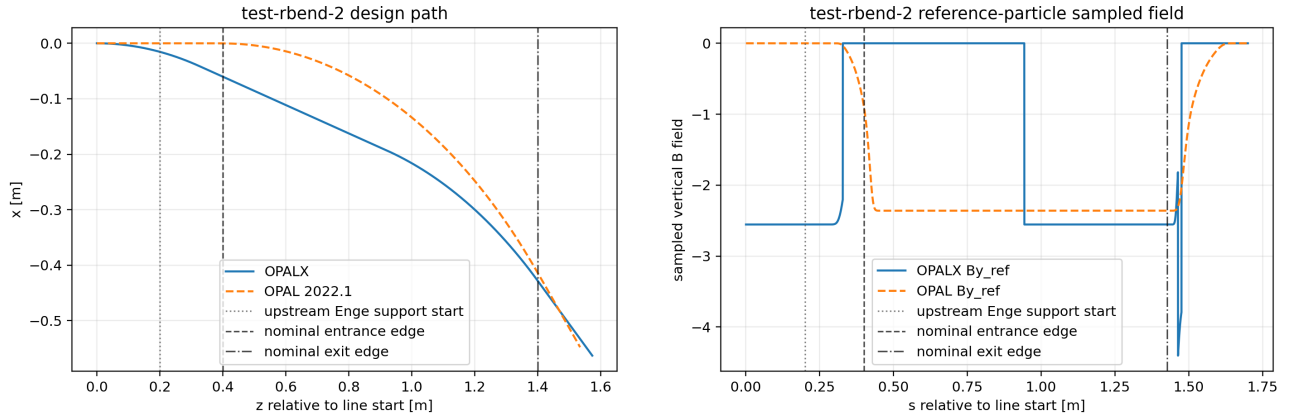


Figure 9: `test-rbend-2` OPALX versus OPAL 2022.1: design-path  $X$ - $Z$  orbit (left) and `.stat` reference  $B_y$  (right).

## 9 Next candidate: symmetric C-shape four-bend chicane

A short web search did not turn up a clean accelerator-school exercise sheet with a worked numerical solution for exactly the OPALX use case. Two school sources are nevertheless immediately useful for

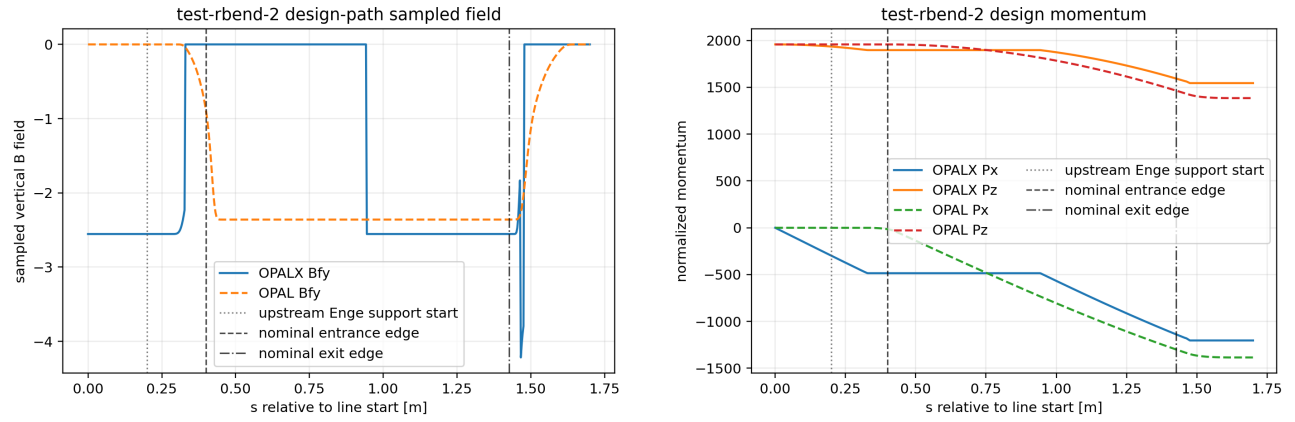


Figure 10: **test-rbend-2** design-path diagnostic comparison. The marker positions use the rectangular RBEND arc length for  $s_{\text{rel}}$  and the straight body length for  $z_{\text{rel}}$ .

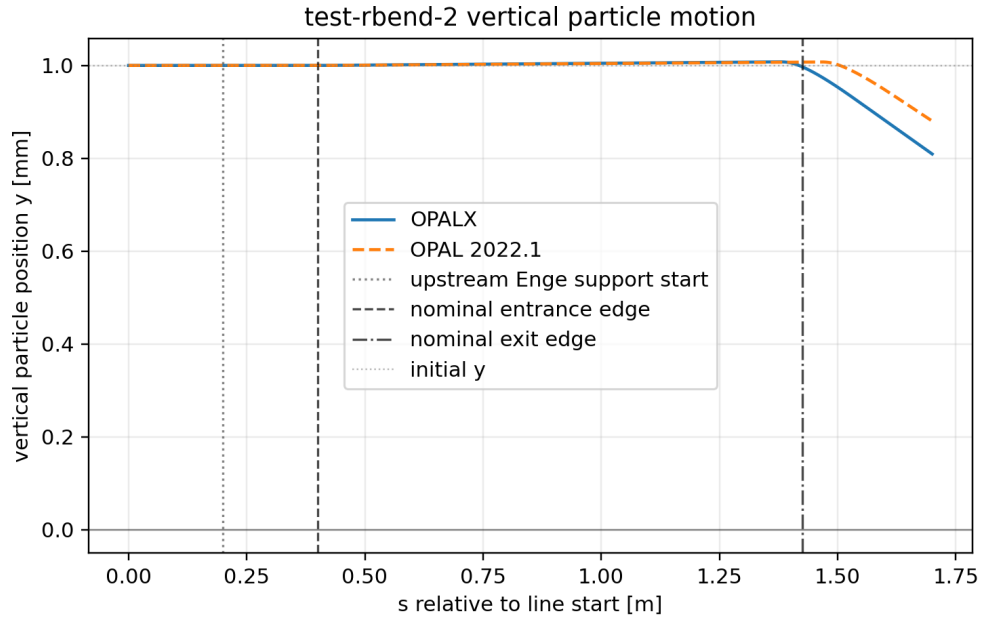


Figure 11: **test-rbend-2** vertical particle motion from the `.stat  $\bar{y}$`  column. The visible vertical difference is measured while the RBEND field magnitude and support still disagree with old OPAL.

a regression design. The USPAS bunch-compression slides give the standard four-rectangular-dipole chicane model and the small-angle expression for  $R_{56}$  [5]. The CAS FEL/ERL proceedings discuss the symmetric C-shape chicane as a common FEL compressor geometry and give the same small-angle model together with the usual higher-order longitudinal coefficients [6].

| source      | useful content                                          | benchmark role          |
|-------------|---------------------------------------------------------|-------------------------|
| USPAS 2014  | four rectangular dipoles, $R_{56}$ , compression map    | primary analytic seed   |
| CAS FEL/ERL | C-shape geometry, $T_{566}$ , $U_{5666}$ , timing model | cross-check and context |

The intended OPALX benchmark is a symmetric C-shape chicane with four equal hard-edge dipoles,

$$(+\theta, -\theta, -\theta, +\theta),$$

magnetic length  $L_b$ , lateral drift  $D$  between dipoles 1–2 and 3–4, and central drift  $C$  between dipoles 2–3. To leading order in  $\theta$ , the central drift does not contribute to the longitudinal dispersion. The school-formula target is therefore

$$R_{56} \simeq -\theta^2 \left( \frac{4}{3}L_b + 2D \right) = -2\theta^2 \left( D + \frac{2}{3}L_b \right).$$

For the same small-angle model one can use

$$T_{566} \simeq -\frac{3}{2}R_{56}, \quad U_{5666} \simeq 2R_{56},$$

as optional higher-order checks once the linear regression is stable.

The executable OPALX realization is

**sandbox/test-chicane-1/test-chicane-1.in.**

It keeps the survey formulas symbolic in the input file. With

$$\rho = \frac{L_b}{2 \sin(\theta/2)},$$

where  $L_b$  is the OPAL/OPALX **SBEND** chord length, not the circular arc length, the reused survey increments are

$$\begin{aligned} \Delta x_b &= \rho(1 - \cos \theta), & \Delta z_b &= \rho \sin \theta, \\ \Delta x_{b/2} &= \rho \left( 1 - \cos \frac{\theta}{2} \right), & \Delta z_{b/2} &= \rho \sin \frac{\theta}{2}, \\ \Delta x_{r/2} &= \rho \left( \cos \frac{\theta}{2} - \cos \theta \right), & \Delta z_{r/2} &= \rho \left( \sin \theta - \sin \frac{\theta}{2} \right), \\ \Delta x_D &= D \sin \theta, & \Delta z_D &= D \cos \theta. \end{aligned}$$

The **REAL** symbols **B1\_X**, **B1\_Z**, ..., **MEXIT\_X**, **MEXIT\_Z** in the input are assembled from these increments and are then used directly in the element **X**, **Y**, **Z**, and **THETA** attributes.

For OPALX the first useful regression should stay purely geometric and single-particle:

1. place the four analytic **SBENDS** explicitly, without field maps;
2. use monitors after each dipole and at the chicane exit;
3. require the final monitor to be parallel to the entrance axis and centered on the design exit;

4. compare the exported element positions with the analytic C-chicane survey.

The second regression should introduce three single-particle runs or three separate particles with

$$\delta = -10^{-3}, \quad 0, \quad +10^{-3},$$

and measure the arrival/path-length shift at the exit monitor. The expected linear fit is

$$\Delta z_f(\delta) - \Delta z_f(0) \simeq R_{56} \delta.$$

This turns the C-chicane into a direct test of both element placing and the time-of-flight/path-length model.

## References

- [1] B. J. Holzer, *Transverse Beam Dynamics*, CERN Yellow Reports: School Proceedings, Vol. 5 (2018), <https://doi.org/10.23730/CYRSP-2018-005.1>.
- [2] K. L. Brown, *A First- and Second-Order Matrix Theory for the Design of Beam Transport Systems and Charged Particle Spectrometers*, SLAC-R-075, Rev. 4, June 1982, <https://www.slac.stanford.edu/pubs/slacreports/reports01/slac-r-075.pdf>.
- [3] H. Grote, F. Schmidt, L. Deniau, and G. Roy, *The MAD-X Program: User's Reference Manual*, CERN, Version 5.02 series, sections on reference-system poses, SBEND, and dipole-edge focusing, <https://madx.web.cern.ch/>.
- [4] D. Sagan, *Bmad Manual*, Cornell CLASSE, sections on SBEND/RBEND, bend coordinates, and fringe-field integrals, <https://www.classe.cornell.edu/bmad/bmad-manual.pdf>.
- [5] D. Nguyen, J. Lewellen, and L. Duffy, *RF Linac for High-Gain FEL: Bunch Compression*, US Particle Accelerator School, June 2014, [https://uspas.fnal.gov/materials/14UNM/E\\_Bunch\\_Compression.pdf](https://uspas.fnal.gov/materials/14UNM/E_Bunch_Compression.pdf).
- [6] S. Di Mitri, *Bunch-Length Compressors*, Proceedings of the CAS-CERN Accelerator School on Free Electron Lasers and Energy Recovery Linacs, CERN Yellow Reports: School Proceedings, Vol. 1 (2018), <https://doi.org/10.23730/CYRSP-2018-001.361>.